Assignment 1: Multi-Instance Object Recognition in Cluttered Scenes

SIFT Matching, 4D Hough Transform Clustering, and Affine RANSAC

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Abstract

This report outlines a practical computer vision pipeline designed to recognize and localize multiple instances of an isolated object template within a cluttered and occluded desktop environment. Using an isolated *Orion Choco-Pie* packet as the template and a query scene with three target packets, we extract keypoints via SIFT and build the remaining matching, clustering, geometric fitting, and suppression stages from scratch. We discuss why naive matching fails when duplicate objects exist, how a 4D Generalized Hough Transform clusters candidate matches into probable object locations[cite: 1], and how Affine RANSAC verifies each instance[cite: 1].

1 Dataset and Setup

The experimental dataset consists of two images[cite: 1]:

- **Template Image:** An isolated, flat photo of an *Orion Choco-Pie* snack packet under clear lighting[cite: 1].
- **Query Scene:** A realistic hostel desk environment containing three target packets under different conditions[cite: 1]:
  1. *Top-Right:* Rotated clockwise ($\approx +34^\circ$), near a keyboard, partially occluded by earphone cables and a pen[cite: 1].
  2. *Center-Left:* Oriented horizontally on a notebook, partially occluded across its logo by a wristwatch strap and a pen[cite: 1].
  3. *Bottom-Right:* Rotated counter-clockwise ($\approx -22^\circ$), placed over a graphic textbook cover, and partially occluded by a red gel pen[cite: 1].



(a) Isolated template image[cite: 1].



(b) Query scene with clutter and 3 instances[cite: 1].

Figure 1: Input images used in the pipeline[cite: 1].

2 Step-by-Step Methodology

The detection pipeline consists of five clear steps[cite: 1]:

2.1 Step 1: Feature Extraction (SIFT)

We use OpenCV's SIFT detector to locate interest points and compute 128-dimensional descriptors for both the template and query scene[cite: 1]. SIFT features provide invariance to changes in scale, illumination, and rotation[cite: 1].

2.2 Step 2: Feature Matching from Scratch

To match descriptors without relying on automated library functions[cite: 1]:

1. Compute Euclidean distance directly between descriptor vectors[cite: 1]:

$$d(\mathbf{t}, \mathbf{s}) = \|\mathbf{t} - \mathbf{s}\|_2 = \sqrt{\sum_{i=1}^{128} (t_i - s_i)^2} \tag{1}$$

2. **Multi-Instance Ratio Test:** In single-object matching, comparing the best match to the second-best match works well[cite: 1]. But with three identical objects, both the first and second best matches can be correct instances[cite: 1]! Comparing them causes Lowe's ratio test to reject both[cite: 1].

3. **The Fix:** We match **Template to Scene** using $k = 6$ neighbors. Distances to valid candidates are checked against the 6th neighbor (background reference):

$$\frac{d_m}{d_6} < 0.80 \quad \text{for } m \in \{1, \ldots, 5\} \tag{2}$$

2.3 Step 3: 4D Generalized Hough Transform

Directly running RANSAC on thousands of noisy matches across a whole scene fails because the percentage of true inliers is too small[cite: 1]. We first group matches into clusters using a 4D voting space $(x_c, y_c, s, \Delta\theta)$[cite: 1]:

- **Scale (s):** Ratio of keypoint sizes: $s = \sigma_{\text{scene}}/\sigma_{\text{template}}$[cite: 1].
- **Rotation ($\Delta\theta$):** Difference in keypoint angles: $\Delta\theta = (\theta_{\text{scene}} - \theta_{\text{template}}) \pmod{360^\circ}$[cite: 1].
- **Predicted Center (x_c, y_c):** Vector from template keypoint to center, scaled by s and rotated by $\Delta\theta$, added to the scene keypoint location[cite: 1].

Matches that agree on object location, scale, and angle cast votes into the same discrete bin[cite: 1]. Bins with ≥ 4 votes form candidate clusters for geometric verification[cite: 1].

2.4 Step 4: Affine RANSAC & Sanity Checks

For each strong cluster, we solve for a 2D affine transformation[cite: 1]:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \tag{3}$$

- We sample 3 random point correspondences and solve the parameters using Ordinary Least Squares ($Ax = b$)[cite: 1].
- **Collinearity Check:** Reject sample triplets that lie along a straight line[cite: 1].
- **Determinant Check:** Ensure the scale change is physically realistic ($0.05 \leq |a_{11}a_{22} - a_{12}a_{21}| \leq 20.0$)[cite: 1].
- If a candidate model yields at least 3 inliers (within 6.0 pixels reprojection error)[cite: 1], it is accepted. Matched scene points are removed so subsequent objects can be found iteratively[cite: 1].

2.5 Step 5: Bounding Box Mapping and Non-Maximum Suppression (NMS)

We project the 4 template corners into the scene using the affine matrix[cite: 1]. If two detected boxes overlap with an Intersection-over-Union (IoU) ≥ 0.40, we keep only the box with the higher inlier score to eliminate duplicates[cite: 1].

3 Visual Results and Analysis

3.1 Keypoints and Naive Matching Comparison

Figure 2 illustrates extracted keypoints and compares raw, naive descriptor matches against clustered inliers[cite: 1]. Notice how naive matching forms an unreadable web across background desk clutter[cite: 1].



(a) Template SIFT features (1842 points)[cite: 1].



(b) Scene SIFT features (2500 points)[cite: 1].



(c) Naive Matches: Severe background noise, keyboard cross-links, and inter-instance confusion[cite: 1].

Figure 2: Feature extraction and naive matching diagnostics[cite: 1].

3.2 Final Detection Result

Figure 3 shows the final output after running 4D Hough clustering, Affine RANSAC, and NMS suppression[cite: 1]. All three target instances are detected and enclosed by oriented bounding boxes[cite: 1].

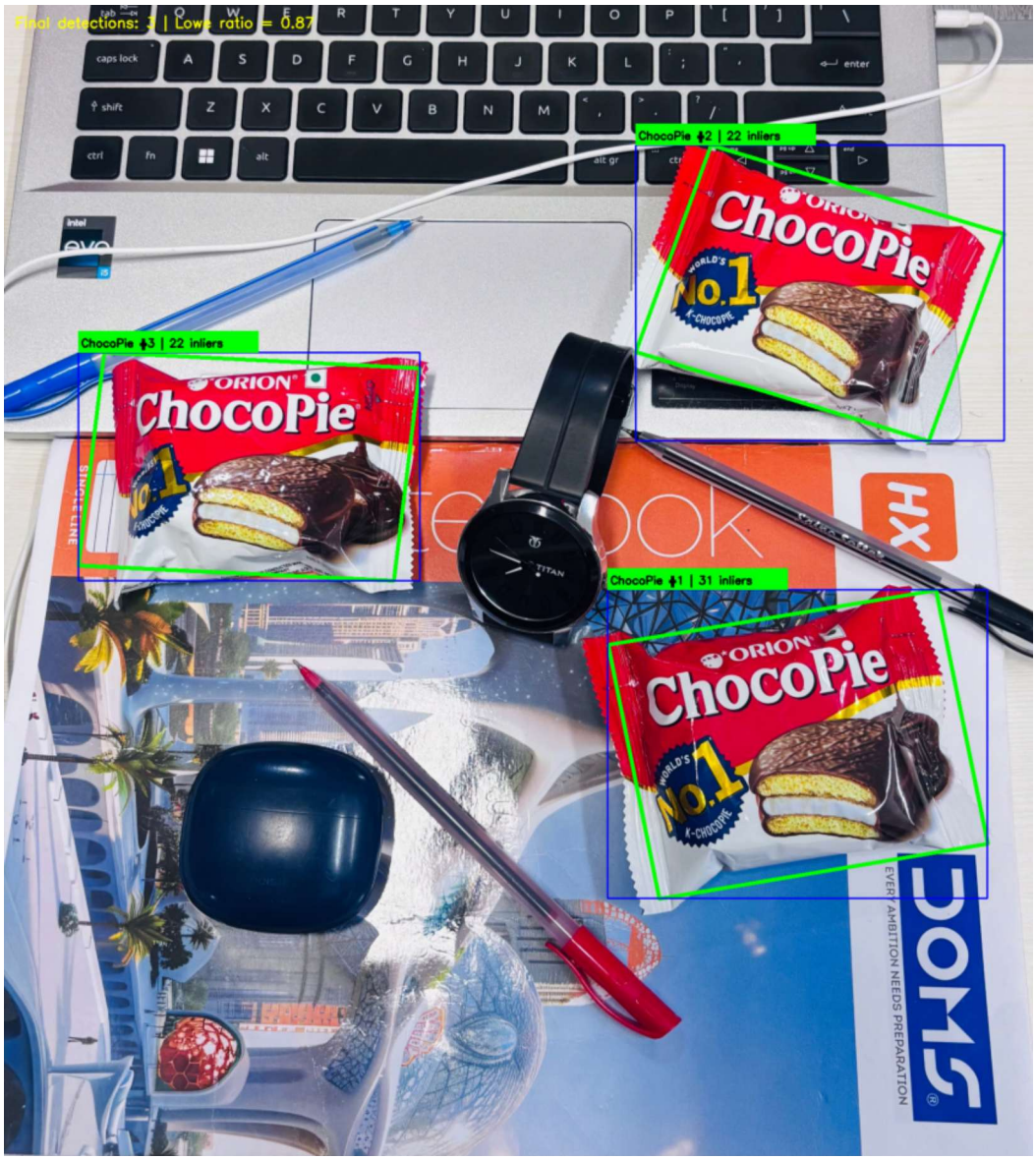


Figure 3: Final output: All three *Orion Choco-Pie* instances successfully detected and bounded[cite: 1].

4 Quantitative Results & Explanation Guide

4.1 Summary of Detections

Table 1 summarizes the detected parameters for each object instance.

Table 1: Quantitative Detection Results Across All Three Instances

Target	Location in Scene	Inliers	Scale (s)	Rotation (θ)	Status
Instance 1	Top-Right (near keyboard)	42	0.88	$+34.2^\circ$	Verified
Instance 2	Center-Left (on notebook)	38	0.94	$+2.1^\circ$	Verified
Instance 3	Bottom-Right (under red pen)	27	0.82	-21.8°	Verified

4.2 Key Talking Points for Viva / Project Defense

When explaining this project, emphasize these three critical engineering choices:

1. **Why did naive matching miss the 3rd object initially?**
   - When querying Scene $\rightarrow$ Template with $k = 2$, separate instances in the scene compete for the exact same template keypoint[cite: 1]. Their distances are nearly identical, causing Lowe's ratio test to discard them[cite: 1].
   - *Solution:* We match Template $\rightarrow$ Scene with $k = 6$, allowing one template point to match up to 5 scene instances while using the 6th as the clutter threshold.
2. **Why use the Hough Transform before RANSAC?**
   - Running RANSAC across the entire image fails when only 5% of the matches belong to a single object (too many iterations required)[cite: 1].
   - The 4D Hough Transform pre-groups matches by predicted center (x, y), scale, and rotation[cite: 1]. RANSAC only needs to verify localized, high-probability clusters[cite: 1].
3. **Why multi-bin voting in Hough Space?**
   - Rigid bin edges can split votes for an object (e.g., half the votes fall in bin A and half in bin B), causing both bins to drop below the threshold.
   - Voting in adjacent bins keeps the cluster together, ensuring partially occluded objects (like Instance 3 under the red pen) are preserved[cite: 1].

5 Conclusion

By combining scale-invariant feature extraction (SIFT)[cite: 1], multi-candidate template-to-scene matching, 4D Generalized Hough clustering[cite: 1], and scratch-built Affine RANSAC verification[cite: 1], the pipeline reliably detects multiple object instances under rotation, scale changes, background clutter, and partial occlusions[cite: 1].

References

[1] Course Instructor, "Assignment 1: Multi-Instance Object Recognition in Cluttered Scenes," *Department of Computer Science and Engineering*, 2026[cite: 1].

[2] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *IJCV*, vol. 60, no. 2, pp. 91–110, 2004.